THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET B

Chapter 8: **Playing with Constructions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A compass keeps the pencil at a fixed distance, called the: 1

(a) diameter **(b)** radius **(c)** chord **(d)** arc

2. The diagonals of a rectangle are: 1

(a) equal in length **(b)** parallel
(c) perpendicular always **(d)** unequal

3. The perpendicular bisector of a segment: 1

(a) is parallel to it **(b)** cuts at one fourth
(c) cuts it at its midpoint at 90° **(d)** makes 60°

4. To draw a line parallel to a given line we can use: 1

(a) a ruler and set square **(b)** a compass only
(c) a balance **(d)** a thermometer

5. The diagonal of a rectangle divides it into: 1

(a) two squares **(b)** four rectangles
(c) two congruent right triangles **(d)** three triangles

6. Which tool helps mark equal lengths accurately? 1

(a) protractor **(b)** thermometer
(c) clock **(d)** compass or divider

7. The number of lines of symmetry of a square is: 1

(a) 2 **(b)** 4 **(c)** 8 **(d)** 1

8. Two circles with centres A and B and equal radii intersect at points that lie on the: 1

(a) tangent **(b)** perpendicular bisector of AB
(c) line AB **(d)** diameter of A

9. A square of sides 7 cm and 7 cm has perimeter: 1

(a) 49 cm **(b)** 14 cm **(c)** 30 cm **(d)** 28 cm

10. A rectangle of sides 5 cm and 3 cm has perimeter: 1

(a) 8 cm **(b)** 16 cm **(c)** 18 cm **(d)** 15 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the instruments in a geometry box used for construction. 2

12. What is the radius of a circle? 2

13. A rectangle has diagonal of 10 cm. What is the length of the other diagonal? 2

14. Why do the diagonals of a rectangle bisect each other? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. State three properties of a square. 3

16. A rectangle has sides 9 cm and 12 cm. The diagonal splits it into two triangles. Find the area of each. 3

17. Describe how to draw the perpendicular bisector of a segment AB of length 7 cm. 3

OR

The diagonals of a rectangle ABCD meet at O. If AC = 14 cm, find OA and BD. 3

18. A tile pattern uses squares of side 10 cm arranged 6 across and 4 down. Find the area and perimeter of the whole rectangle. 3

OR

Why can a ruler and compass help to draw a perpendicular at a point on a line? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Explain the properties of diagonals of rectangles and squares.
(ii) Compare them in a table. 5

OR

(i) Draw a circle of radius 4 cm.
(ii) Draw two chords.
(iii) Construct the perpendicular bisector of each.
(iv) What do you observe? 5

20. **Case study:** Two villages A and B are 6 km apart. A well is to be dug equidistant from both.
(i) Where can it be placed?
(ii) Draw the possible positions.
(iii) Which point on AB is best?
(iv) How far from each? 5

OR

(i) Using ruler and compass, copy a line segment of 5.5 cm.
(ii) Draw a square on it.
(iii) Name its parts. 5

END OF QUESTION PAPER — Total: 40 Marks

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